Emotion Detection in Multilingual Social Media Text Using NLP

Darshil Maniya – 12302130601012
Department of Computer Science And Design, A.D. Patel Institute of Technology, Anand, Gujarat, India

***Abstract*—This paper studies multi-label emotion detection (anger, fear, joy, sadness and surprise) in social media text written in English, Hindi and German. Using the BRIGHTER corpus (7,130 training, 793 validation and 830 held-out posts), a language-independent normalization step feeds three models: a word-level TF–IDF Logistic Regression baseline, a word-plus-character TF–IDF Logistic Regression, and a fine-tuned XLM-RoBERTa-base with a sigmoid output layer. Decision thresholds are tuned per emotion on validation data only. On the held-out set the baseline reaches 0.5612 macro-F1; character n-grams raise it to 0.6027 (paired bootstrap difference +0.041, 95% interval [+0.025, +0.059]); and XLM-RoBERTa reaches 0.6741 (4 epochs) and 0.6819 (5 epochs), while subset accuracy rises from 0.393 to 0.434 and 0.542. Error analysis of the TF–IDF model shows more false positives (456) than false negatives (282) and a language gap: the any-label error rate is 0.79 for English, 0.55 for German and 0.35 for Hindi, which follows the number of emotions per post. The results come from single runs on a small held-out set, the XLM-RoBERTa gains carry no confidence interval, and the two training budgets differ by less than the evidence can resolve; the system is not a tool for decisions about individuals.**

***Index Terms*—emotion detection, multilingual NLP, social media text, multi-label classification, TF–IDF, XLM-RoBERTa, error analysis**

I. INTRODUCTION

Emotion detection assigns one or more emotion categories to a short text and is usually treated as supervised text classification. On social media it is difficult because posts are short, informal and full of slang, emojis and spelling variation, and because people express emotion differently across languages and cultures. Most emotion resources and models target English, which leaves speakers of other languages poorly served. Resources such as the BRIGHTER corpus [1] and the SemEval-2025 Task 11 shared task [2] provide human-annotated emotion data in many languages and make a systematic multilingual comparison possible.

This study targets multi-label detection of five emotions, anger, fear, joy, sadness and surprise, a label set in the basic-emotion tradition [3]. A post may carry several emotions or none. We work with three languages that differ in script, resource level and text style: English, Hindi and German. The pipeline normalizes social media text without language-specific stopword removal or stemming, builds word and character TF–IDF features, and compares sparse linear classifiers with a fine-tuned multilingual transformer, XLM-RoBERTa. Source code, result files and dataset statistics are kept with the project so that the experiments can be repeated.

The contributions are: 1) a comparison of a word-level TF–IDF Logistic Regression baseline, a word-plus-character TF–IDF Logistic Regression and a fine-tuned XLM-RoBERTa under one fixed protocol on three languages; 2) a description of corpus size, label distribution, multi-label share and text length per language; and 3) an evaluation with validation-only threshold tuning, a paired bootstrap interval for the TF–IDF comparison, per-emotion and per-language error analysis, and a training-budget ablation for XLM-RoBERTa.

II. RELATED WORK

Bag-of-words features and linear classifiers remain strong, efficient baselines for text categorization; Pang, Lee and Vaithyanathan framed sentiment analysis as supervised classification with such features [4]. TF–IDF weighting down-weights terms that occur in many documents [5], and logistic regression estimates class probabilities from a linear combination of features [6]. Character n-grams are attractive for multilingual and noisy text because they need no language-specific tokenizer. The scikit-learn library provides the vectorization, model fitting and evaluation used here [7].

Emotion corpora have been dominated by English. CARER provides English tweets labelled from emotion hashtags [8], SemEval-2018 Task 1 added Arabic and Spanish tweets with multi-label annotation [9], GoEmotions offers fine-grained English Reddit comments [10], and XED covers many languages through annotation projection [11]. The BRIGHTER corpus [1] and SemEval-2025 Task 11 [2] extend human-annotated, multi-label emotion data to a much wider set of languages, including several low-resource ones; this study uses its English, Hindi and German portions.

Pretrained transformers such as BERT [12] model context and negation that sparse n-grams cannot. XLM-RoBERTa [13] is pretrained on text in about 100 languages and transfers across them, and the Hugging Face Transformers library makes fine-tuning straightforward [14]. Variants adapted to Twitter [15] or to Indian languages [16] exist but are not evaluated here. The trade-off is cost: transformers need a GPU and are harder to inspect than sparse linear models, which motivates comparing both families on the same data.

III. MODEL AND DATA

A. Task Definition

Given a post x, the classifier predicts a binary vector $y \in \{0,1\}^5$, with one entry for each emotion in {anger, fear, joy, sadness, surprise}. The emotions are not mutually exclusive, and the all-zero vector means that the post expresses none of the five emotions. The language of the post is not given to the model as a feature.

B. Dataset and Corpus Description

We use the BRIGHTER corpus [1] through its Hugging Face release (brighter-dataset/BRIGHTER-emotion-categories), configurations eng, hin and deu. Each configuration has train, dev and test splits. The official train splits contain 2,764 (English), 2,556 (Hindi) and 2,603 (German) posts. We hold out a language-stratified 10% of the pooled training data as validation data and report results on the official dev splits (230, 200 and 400 posts). The official test splits (5,528, 2,020 and 5,208 rows) are not used in this study. Table I summarizes the data.

TABLE I
DATASET SUMMARY

Item	Value
Corpus	BRIGHTER (SemEval-2025 Task 11, Track A) [1], [2]
Languages used	English, Hindi, German
Train / val. / held-out, English	2,487 / 277 / 230
Train / val. / held-out, Hindi	2,300 / 256 / 200
Train / val. / held-out, German	2,343 / 260 / 400
Total train / val. / held-out	7,130 / 793 / 830
Labels	anger, fear, joy, sadness, surprise (multi-label)
Avg. length, words (train): En / Hi / De	15.4 / 16.7 / 34.8
Training posts per label (anger, fear, joy, sadness, surprise)	1,376 / 2,015 / 1,511 / 1,650 / 1,135
Largest / smallest label share	28.3% (fear) / 15.9% (surprise)
No-emotion posts (train): En / Hi / De	8.8% / 29.3% / 30.9%
Multi-label posts (train): En / Hi / De	49.9% / 7.0% / 16.2%

Held-out = official dev splits. Label shares are the percentage of the 7,130 training posts carrying the label.

The three languages differ markedly. English posts average 15.4 words, Hindi posts 16.7 and German posts 34.8, so German texts are more than twice as long, which suggests a different text source that was not audited here. English is the most label-dense: 49.9% of its training posts carry two or more emotions, fear appears in 58.3% of them, and only 8.8% carry no emotion. In Hindi, 7.0% of posts are multi-label and 29.3% carry no emotion; in German the shares are 16.2% and 30.9%. Pooled over languages, fear is the most frequent label and surprise the least frequent (Fig. 1). The held-out set has 830 posts, so each emotion has between 110 and 198 positive examples. A duplicate and train–test overlap audit was not performed.

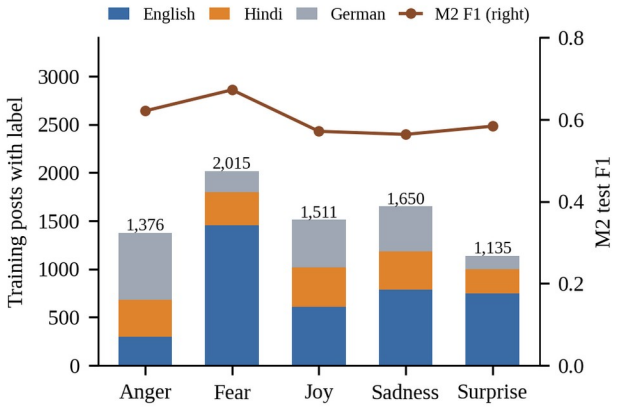


Fig. 1. Training posts per emotion, stacked by language (bars, left axis), and test F1 per emotion of model M2 (line, right axis).

C. Text Normalization

Each post is Unicode NFKC-normalized; URLs are replaced by a <url> token and @mentions by a <user> token; the # symbol is removed and the hashtag word is kept; characters repeated three or more times are reduced to two (sooooo becomes soo); text is lowercased and whitespace is collapsed. Emojis and punctuation are kept. No stopword removal, stemming or lemmatization is applied, because such resources are language-specific and because removing function words such as not or never would discard negation. The same normalized text is given to the TF–IDF models and to the XLM-RoBERTa tokenizer.

D. Feature Representation

Model M1 uses TF–IDF over word unigrams and bigrams, with a token pattern that keeps single characters and punctuation or emoji symbols as tokens, a minimum document frequency of 2 and sublinear term frequency. For a term t and document d,

$$tfidf(t,d) = (1 + \ln f_{t,d}) \cdot idf(t) \quad (1)$$

$$idf(t) = \ln((1 + N) / (1 + df(t))) + 1 \quad (2)$$

where $f_{t,d}$ is the term count, N the number of training documents and $df(t)$ the number of training documents containing t; each document vector is then L2-normalized. Model M2 concatenates this word block with a second TF–IDF block over character n-grams of length 2 to 5 inside word boundaries (minimum document frequency 3, at most 300,000 features), which tolerates spelling variation and inflection and needs no tokenizer. Vocabularies and IDF weights are fitted on training data only. Model M3 uses no hand-built features: XLM-RoBERTa splits the text into SentencePiece subwords from a vocabulary shared by about 100 languages and learns contextual representations during fine-tuning.

E. Classifiers

Baseline (M1). One-vs-rest Logistic Regression on the word TF–IDF features. For each emotion j,

$$P(y_j = 1 \mid x) = \sigma(w_j \cdot x + b_j) \quad (3)$$

and training minimizes the class-weighted logistic loss with L2 regularization,

$$L_j = \tfrac{1}{2}\|w_j\|^2 + C \sum_i c_{ij} \ln(1 + \exp(-s_{ij} z_{ij})) \quad (4)$$

where $z_{ij} = w_j \cdot x_i + b_j$, $s_{ij} = +1$ if $y_{ij} = 1$ and -1 otherwise, and $c_{ij} = n / (2n_{j,y})$ is the balanced class weight, with $n_{j,y}$ the number of training posts whose label for emotion j equals y. We use the liblinear solver with $C = 4.0$, fixed in advance and not tuned.

Improved classical model (M2). The same classifier on the concatenated word and character TF–IDF features.

Transformer (M3). The xlm-roberta-base model [13] with the standard sequence-classification head of Transformers, which maps the first-token representation to five logits z_{ij}. With sigmoid outputs $p_{ij} = \sigma(z_{ij})$, training minimizes a positively weighted binary cross-entropy averaged over posts and emotions,

$$L = -\sum_{i,j} [\omega_j y_{ij} \ln p_{ij} + (1 - y_{ij}) \ln(1 - p_{ij})] \quad (5)$$

with $\omega_j = \sqrt{(n_j^- / n_j^+)}$, the square root of the ratio of negative to positive training posts for emotion j. Training uses AdamW [17] (learning rate 2×10^{-5}, weight decay 0.01), batch size 16, a maximum of 128 subwords, linear warm-up over the first 10% of steps followed by linear decay, gradient-norm clipping at 1.0 and fp16 mixed precision on a GPU. We train for 4 epochs (default) and for 5 epochs, keep the epoch with the highest validation macro-F1 at threshold 0.5, and use its predictions.

Decision rule. For all models, emotion j is predicted when

$$\hat{y}_j = 1[\, p_j \geq \tau_j \,] \quad (6)$$

where τ_j is chosen from {0.10, 0.15, …, 0.90} to maximize the validation F1 of emotion j and is then frozen for the held-out set.

IV. EXPERIMENTAL PROTOCOL

The experiment uses a language-stratified 90/10 split of the official training data for training and validation (seed 42) and the official dev splits as the held-out test set. The validation set is used only to select the XLM-RoBERTa epoch and the per-emotion thresholds. All TF–IDF vocabularies, IDF weights and class weights are fitted on the training posts only. All models are trained on the pooled three-language training data and evaluated on the pooled held-out set; per-language results are reported for M2 only.

Following multi-label practice [18], we report micro- and macro-averaged precision, recall and F1; subset accuracy, the share of posts whose whole label vector is exactly correct; and Hamming loss, the share of incorrect label decisions (lower is better). Macro-F1 is the primary metric because it weights every emotion equally. Two trivial references computed from the held-out label counts are included: predicting no emotion for every post, and predicting all five emotions for every post.

A paired bootstrap [19] over the 830 held-out posts (1,000 resamples) gives a 95% interval for the macro-F1 difference between M2 and M1. For error analysis, the held-out predictions of M2, the best TF–IDF model by macro-F1, are used to count false positives and false negatives per emotion and to compute error rates per language and per text-length quartile; XLM-RoBERTa predictions were not analysed in the same way. Software: Python with scikit-learn [7] and PyTorch with Hugging Face Transformers [14], run on a GPU in Google Colab. Each configuration was run once with seed 42; repeated seeds, cross-validation and confidence intervals for XLM-RoBERTa are not used.

V. RESULTS

A. Overall Performance

TABLE II
OVERALL HELD-OUT PERFORMANCE

Model	Micro F1	Macro P	Macro R	Macro F1	Subset acc.	Ham. loss
Predict none (ref.)	0.0000	–	–	0.0000	0.2627	0.2048
Predict all (ref.)	0.3400	0.2048	1.0000	0.3382	–	0.7952
M1: TF–IDF word + LR	0.5690	0.5158	0.6346	0.5612	0.3928	0.1957
M2: TF–IDF word+char + LR	0.6062	0.5594	0.6690	0.6027	0.4337	0.1778
M3: XLM-R, 4 epochs	0.6764	–	–	0.6741	0.5181	0.1383
M3: XLM-R, 5 epochs	**0.6859**	–	–	**0.6819**	**0.5422**	**0.1263**

Held-out set, 830 posts. LR = Logistic Regression; Ham. loss = Hamming loss. References are computed from the held-out label counts. "–" = not defined, not logged or not computed. Bold = best.

Table II summarizes the held-out results. Predicting no emotion gives a Hamming loss of 0.2048 and a subset accuracy of 0.2627, because 218 of the 830 posts carry no label; predicting all emotions gives a macro-F1 of 0.3382. Every trained model clearly exceeds these references in F1. The baseline M1 reaches a macro-F1 of 0.5612, with higher recall (0.6346) than precision (0.5158). Adding character n-grams (M2) raises macro-F1 to 0.6027 (+0.0415), micro-F1 to 0.6062 and subset accuracy to 0.4337. The paired bootstrap difference in macro-F1 is +0.0413 with a 95% interval of [+0.0254, +0.0590], so on this held-out set the gain is unlikely to be a resampling artefact.

Fine-tuned XLM-RoBERTa is better again: macro-F1 is 0.6741 after 4 epochs and 0.6819 after 5, which is +0.0714 and +0.0792 over M2 and +0.1129 and +0.1207 over M1. Subset accuracy reaches 0.5181 and 0.5422, and Hamming loss falls from 0.1778 (M2) to 0.1383 and 0.1263. Macro precision and recall of XLM-RoBERTa were not logged and are not reported, and no bootstrap interval was computed for its comparison with M2. Hamming loss of M1 (0.1957) is only slightly below the predict-none reference (0.2048), which shows that Hamming loss rewards predicting few labels when labels are sparse; macro-F1 and subset accuracy are more informative here. Fig. 2 plots the main scores.

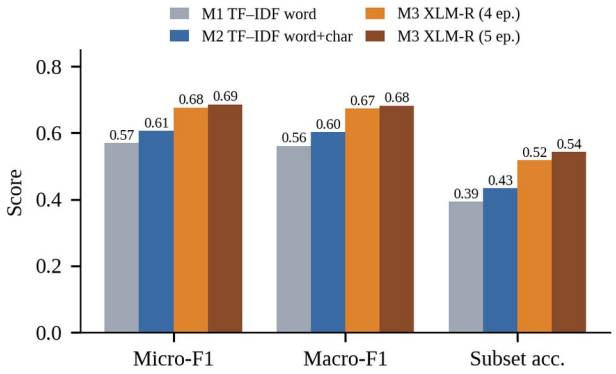


Fig. 2. Held-out micro-F1, macro-F1 and subset accuracy of the four trained configurations.

B. Per-Emotion Performance

TABLE III
PER-EMOTION HELD-OUT RESULTS, M2

Emotion	Supp.	TP	FP	FN	P	R	F1
Anger	186	128	98	58	0.566	0.688	0.621
Fear	192	148	100	44	0.597	0.771	0.673
Joy	164	84	46	80	0.646	0.512	0.571
Sadness	198	128	128	70	0.500	0.646	0.564
Surprise	110	80	84	30	0.488	0.727	0.584
Total / macro	**850**	**568**	**456**	**282**	**0.559**	**0.669**	**0.603**

TP = Supp. − FN. P, R and F1 are derived from the support and error counts; their macro averages reproduce the logged macro precision, recall and F1.

Table III gives per-emotion counts and scores for M2 (Fig. 3). F1 ranges from 0.564 (sadness) to 0.673 (fear). Fear has the highest recall (0.771) and F1. Sadness has the lowest F1 because of low precision (0.500, with 128 false positives), and surprise has the lowest precision (0.488). Joy is the only emotion that is missed more often than it is wrongly predicted (80 false negatives against 46 false positives, recall 0.512). Overall, M2 makes 456 false positives and 282 false negatives, so it over-predicts emotions. This pattern is consistent with class-weighted training and F1-oriented thresholds, which both favour recall, but no ablation without threshold tuning was run to confirm it. Per-emotion scores for XLM-RoBERTa were not analysed.

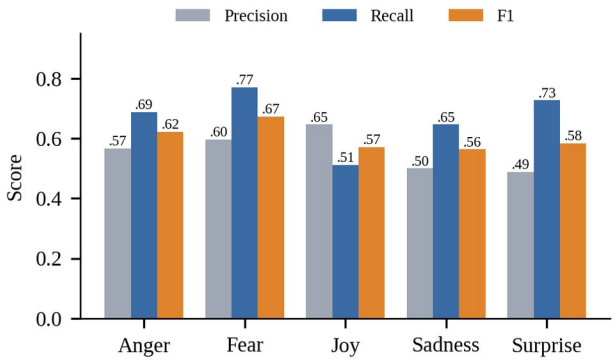


Fig. 3. Held-out precision, recall and F1 per emotion for M2.

C. Error Analysis by Language and Text Length

TABLE IV
M2 HELD-OUT ERRORS BY LANGUAGE

Lang.	Posts	Multi-label %	No-emotion %	Any-label error	Ham. (M2)	Ham. (none)
English	230	48.7	11.3	0.791	0.296	0.303
German	400	18.0	29.5	0.545	0.155	0.184
Hindi	200	4.0	37.0	0.350	0.088	0.134
All	**830**	**–**	**–**	**0.566**	**0.178**	**0.205**

Any-label error = share of posts with at least one wrong label. Ham. = Hamming loss; (none) = predicting no emotion.

Table IV splits the M2 errors by language (Fig. 4). The any-label error rate is 0.791 for English, 0.545 for German and 0.350 for Hindi, and the Hamming loss is 0.296, 0.155 and 0.088, although the three languages contribute similar amounts of training data (2,487, 2,343 and 2,300 posts). The English held-out posts also carry the most labels per post (1.51, against 0.92 for German and 0.67 for Hindi), and 48.7% of them are multi-label, against 4.0% for Hindi. With more labels per post there are more ways to get a label vector wrong. For English the Hamming loss of M2 (0.296) is almost the same as that of predicting no emotion at all (0.303), whereas for German (0.155 against 0.184) and Hindi (0.088 against 0.134) the model is clearly better than that reference. The ordering of the languages therefore matches label density and multi-label share, but language, text source, annotation practice and length are confounded in this corpus, so the gap cannot be attributed to any single factor.

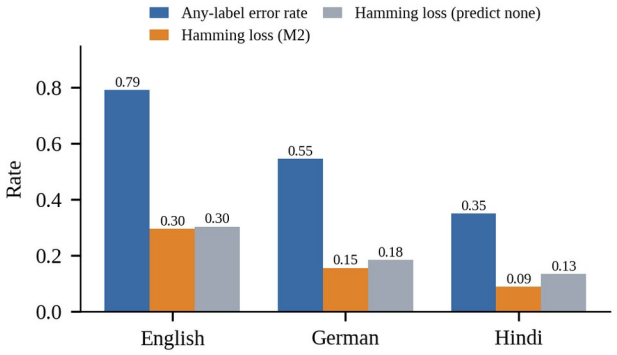


Fig. 4. M2 held-out error by language: any-label error rate, Hamming loss and the Hamming loss of predicting no emotion.

By text length, the any-label error rate of M2 is 0.574, 0.519, 0.587 and 0.586 for the four word-count quartiles (2–11, 12–18, 19–30 and 31–111 words; 216, 208, 208 and 198 posts), with no monotonic trend. Length is itself strongly tied to language, since German posts are the longest, so this result should not be read as a pure length effect.

D. Training-Budget Ablation

TABLE V
XLM-ROBERTA: 4 VERSUS 5 TRAINING EPOCHS

Setting	Sel. epoch	Val. macro F1	Test macro F1	Test micro F1	Subset acc.	Ham. loss
4 epochs	4	0.7184	0.6741	0.6764	0.5181	0.1383
5 epochs	4	0.7100	0.6819	0.6859	0.5422	0.1263
Difference (5 − 4)	**–**	**−0.0084**	**+0.0078**	**+0.0095**	**+0.0241**	**−0.0120**

Validation macro-F1 is measured at threshold 0.5; test scores use tuned thresholds.

Table V compares the two XLM-RoBERTa runs, which differ only in the number of training epochs, so that the linear learning-rate schedule is stretched over a longer horizon; warm-up, gradient clipping and mixed precision were part of both runs. The 5-epoch run scores +0.0078 macro-F1, +0.0095 micro-F1 and +2.41 points of subset accuracy. The evidence does not support concluding that more epochs help. Validation macro-F1 selected epoch 4 in both runs, and it was higher for the 4-epoch run (0.7184 against 0.7100), so the reported 5-epoch model is its fourth epoch, trained under a longer schedule, and the fifth epoch was never selected. With one seed per setting and 830 held-out posts, a difference of 0.008 macro-F1 may well lie within seed-to-seed variation. Validation macro-F1 at threshold 0.5 (0.71–0.72) is also higher than the held-out macro-F1 (0.67–0.68) even though the held-out scores use tuned thresholds; the validation posts come from the language-balanced training split, whereas the held-out set has a different language mix (48% German), so the two are not directly comparable.

VI. DISCUSSION AND LIMITATIONS

Under one protocol, macro-F1, micro-F1, subset accuracy and Hamming loss all improve from word TF–IDF (macro-F1 0.5612) to word-plus-character TF–IDF (0.6027) to fine-tuned XLM-RoBERTa (0.6741–0.6819). Character n-grams add no language-specific resources and still give a consistent gain over words alone, and a multilingual transformer gives a larger one, most visibly in exact-match accuracy (0.393, 0.434, then 0.518–0.542). Because labels are sparse and unevenly distributed across languages, applications should monitor macro-F1 and subset accuracy rather than Hamming loss alone.

The results have several limits. First, the held-out set is the official dev split with only 830 posts (400 German, 230 English, 200 Hindi) and 110–198 positive examples per emotion; the official test splits were not used, and per-language results exist for M2 only. Second, each configuration was run once, so seed-to-seed variability is unknown, XLM-RoBERTa has no confidence interval, and the 0.0078 gap between its two budgets should not be treated as real. Third, the error analysis covers M2 only: XLM-RoBERTa errors by emotion and by language, and a qualitative review of individual errors, are missing, and the link between the language gap and label density is an association in confounded data. Fourth, no duplicate or train–test overlap audit was performed, annotation agreement is not examined here, and the text source of each language was not audited. Fifth, only three languages were studied, none of them low-resource, and code-mixed or transliterated text was not examined; because models were trained on pooled data, the benefit each language draws from the others is unknown, as monolingual training and zero-shot transfer were not run. Sixth, normalization lowercases text for XLM-RoBERTa although its pretraining is case-sensitive, no normalization ablation was run, and hyperparameters were fixed rather than tuned. Seventh, a five-label multi-label scheme cannot represent finer emotions or intensity. Finally, emotion predictions on social media text are noisy and must not be used for decisions about individuals, profiling or surveillance.

Future work should repeat the experiments with several seeds or cross-validation and report confidence intervals, including for XLM-RoBERTa; evaluate on the official test splits; analyse XLM-RoBERTa errors per emotion and language and categorize a sample of errors manually (sarcasm, negation, missing context, ambiguous labels); run monolingual training and zero-shot transfer from English to Hindi and German; try Twitter-adapted or Indian-language models [15], [16] and an ensemble of TF–IDF and XLM-RoBERTa probabilities; and extend the study to low-resource and code-mixed languages.

VII. CONCLUSION

This paper compared a word-level TF–IDF Logistic Regression baseline, a word-plus-character TF–IDF model and fine-tuned XLM-RoBERTa for multi-label emotion detection in English, Hindi and German posts from the BRIGHTER corpus. On the 830-post held-out set, macro-F1 rose from 0.5612 to 0.6027 with character n-grams (paired bootstrap 95% interval for the gain [+0.025, +0.059]) and to 0.6741–0.6819 with XLM-RoBERTa, while subset accuracy rose from 0.393 to as much as 0.542. Error analysis of the TF–IDF model showed more false positives than false negatives and the highest error rate in English, where posts carry the most emotions. These results come from single runs on a small held-out set and do not establish performance on other languages or domains, and the difference between 4 and 5 XLM-RoBERTa epochs is not reliable. Repeated runs, evaluation on the test splits, XLM-RoBERTa error analysis and zero-shot transfer are the natural next steps.

REFERENCES

[1] S. H. Muhammad et al., “BRIGHTER: BRIdging the Gap in Human-annotated Textual Emotion Recognition Datasets for 28 Languages,” arXiv preprint, 2025.

[2] S. H. Muhammad et al., “SemEval-2025 Task 11: Bridging the gap in text-based emotion detection,” in Proc. Int. Workshop on Semantic Evaluation (SemEval), 2025.

[3] P. Ekman, “An argument for basic emotions,” Cognition & Emotion, vol. 6, no. 3–4, pp. 169–200, 1992.

[4] B. Pang, L. Lee, and S. Vaithyanathan, “Thumbs up? Sentiment classification using machine learning techniques,” in Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP), 2002, pp. 79–86.

[5] K. Spärck Jones, “A statistical interpretation of term specificity and its application in retrieval,” J. Documentation, vol. 28, no. 1, pp. 11–21, 1972.

[6] C. M. Bishop, Pattern Recognition and Machine Learning. New York, NY, USA: Springer, 2006.

[7] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” J. Mach. Learn. Res., vol. 12, pp. 2825–2830, 2011.

[8] E. Saravia, H.-C. T. Liu, Y.-H. Huang, J. Wu, and Y.-S. Chen, “CARER: Contextualized affect representations for emotion recognition,” in Proc. EMNLP, 2018, pp. 3687–3697.

[9] S. Mohammad, F. Bravo-Marquez, M. Salameh, and S. Kiritchenko, “SemEval-2018 Task 1: Affect in tweets,” in Proc. Int. Workshop on Semantic Evaluation (SemEval), 2018, pp. 1–17.

[10] D. Demszky, D. Movshovitz-Attias, J. Ko, A. Cowen, G. Nemade, and S. Ravi, “GoEmotions: A dataset of fine-grained emotions,” in Proc. Annu. Meeting Assoc. Computational Linguistics (ACL), 2020, pp. 4040–4054.

[11] E. Öhman, M. Pàmies, K. Kajava, and J. Tiedemann, “XED: A multilingual dataset for sentiment analysis and emotion detection,” in Proc. Int. Conf. Computational Linguistics (COLING), 2020.

[12] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in Proc. NAACL-HLT, 2019, pp. 4171–4186.

[13] A. Conneau et al., “Unsupervised cross-lingual representation learning at scale,” in Proc. ACL, 2020, pp. 8440–8451.

[14] T. Wolf et al., “Transformers: State-of-the-art natural language processing,” in Proc. EMNLP: System Demonstrations, 2020, pp. 38–45.

[15] F. Barbieri, L. Espinosa Anke, and J. Camacho-Collados, “XLM-T: Multilingual language models in Twitter for sentiment analysis and beyond,” in Proc. Language Resources and Evaluation Conf. (LREC), 2022.

[16] S. Khanuja et al., “MuRIL: Multilingual representations for Indian languages,” arXiv preprint, 2021.

[17] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in Proc. Int. Conf. Learning Representations (ICLR), 2019.

[18] M.-L. Zhang and Z.-H. Zhou, “A review on multi-label learning algorithms,” IEEE Trans. Knowl. Data Eng., vol. 26, no. 8, pp. 1819–1837, 2014.

[19] B. Efron and R. J. Tibshirani, An Introduction to the Bootstrap. New York, NY, USA: Chapman & Hall, 1993.